

AVISHKAR MAHESH PAWAR

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Professional Summary

Results-driven Computer Science student with expertise in backend development and distributed infrastructure. Built scalable message queue systems and RESTful APIs using Java, Spring Boot, and Node.js. Experience with high-throughput systems processing 50,000+ messages/sec, consumer group coordination, and microservices architecture. Strong foundation in system design, performance optimization, and cloud technologies. Currently working as Software Developer intern at OSMOS.

Education

MIT Academy of Engineering, Pune

2022 - 2026

B.Tech. - Computer Engineering — CGPA: 9.24/10

Projects

Event Streaming Platform

Java 17, Spring Boot, Netty, Apache Kafka Protocol

- Architected a distributed message queue system processing 50,000+ messages/sec with guaranteed ordering and at-least-once delivery semantics
- Implemented log-based storage with memory-mapped files and zero-copy transfer, achieving 3x throughput improvement over traditional I/O
- Built consumer group coordinator with automatic partition rebalancing, supporting 100+ concurrent consumers with <100ms rebalance time

MiniDB — Distributed Database with ACID Transactions

C++17, Boost.Asio, Protocol Buffers, CMake

- Engineered a fully-featured distributed key-value database from scratch in C++17 (all 6 phases complete) supporting ACID transactions with MVCC concurrency control, handling 10,000+ ops/sec with <1ms latency
- Implemented B+ Tree indexing with LRU-based buffer pool management, reducing disk I/O by 60% through intelligent page caching and prefetching
- Built complete query processing engine with SQL-like parser supporting JOINS, and designed custom binary protocol over TCP using Boost.Asio with connection pooling

AI-Powered Vector Database

Python 3.11, FastAPI, NumPy, Numba

- Implemented graph-based indexing (HNSW) with hierarchical layers for sub-linear search complexity $O(\log n)$ on 1,000,000+ vectors
- Designed sharding mechanism with consistent hashing and write-ahead logging for horizontal scalability and crash recovery
- Built RESTful API using FastAPI with async endpoints handling 1000+ concurrent requests

Technical Skills

Languages: C++, Java, Python, JavaScript, TypeScript

Frameworks: Spring Boot, FastAPI, React, Node.js, Express.js

AI/ML: TensorFlow, PyTorch, OpenAI API, LangChain, NumPy, Numba

Databases: PostgreSQL, MySQL, MongoDB, Redis, Custom Database Implementations

Tools: Git, Docker, Linux, AWS, CMake, Maven, Prometheus

Distributed Systems: Kafka, gRPC, Protocol Buffers, Raft Consensus, Consistent Hashing

CS Fundamentals: Data Structures & Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Systems, System Design

Experience

OSMOS

Jul 2025 - Present

Software Developer

- Worked on backend services using Node.js and developed applications using Python
- Contributed to product development from scratch, including designing, implementing, and testing core features

Core-Decimal Solutions

Feb 2025 - Aug 2025

Software Developer

- Contributed to full-stack project using Vue.js, Express.js, and Sequelize with MVC architecture
- Developed reusable Vue.js components and integrated RESTful APIs with Axios

Achievements & Certifications

GATE 2026: Secured AIR 3124 with GATE Score 601 and 51.45/100 marks in CS — 3x rank improvement from GATE 2025 (AIR 9860)

GATE 2025: Qualified with AIR 9860, GATE Score: 453, Marks: 39.32/100

First Place in Technodium 25: Won first place in technical competition

Certifications: CCNA: Introduction to Networks, CCNA: Switching, Routing, and Wireless Essentials, CCNA: Enterprise Networking, Security, and Automation

Extra-Curricular Activities

Outdoor Sports & Games: Active participant in outdoor sports and recreational activities